

Carlo Makeever

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Professional Summary

Learning is how I give back to the universe; building is how I give back to myself. I'm an aspiring systems engineer, I enjoy the challenge of translating big-picture objectives into measurable requirements. My background blending physics and CS means I think of problem solving in the full chain. The goal is simple: dependable instruments and software that make new science possible and understandable.

Education

University of California - Santa Barbara

Bachelor's, Computer Science

- System Design/AI & ML/Data Analysis Track
- Computer Science Club
- Hackathon Club

September 2025 - May 2027

GPA: 3.9

University of California - Santa Barbara

Bachelor's, Physics

- Rocket Club
- Physics Club

September 2025 - May 2027

GPA: 3.9

San Diego Mesa College

Associate's, Computer Science

- Peer Advisor & Mentor
- Computer Science Club
- STEM Club
- Robotics Club

August 2022 - May 2025

GPA: 3.88

Professional Experience

NASA Jet Propulsion Laboratory (JPL)

Thermal Software and Data Engineer

Los Angeles, CA, USA

May 2025 - September 2025

- Engineered a full-stack data system for the Mars Science Laboratory (Curiosity Rover) thermal subsystem, integrating backend pipelines, Elasticsearch indices, and a frontend dashboard for tactical and strategic analysis
- Designed optimized schemas and ingestion workflows to transform raw rover telemetry into structured, query-ready datasets, reducing analysis latency and enabling multi-SOL trend detection
- Built Python frameworks for ingestion, XML parsing, and margin verification, integrated with automated validation routines to enforce mission safety constraints
- Deployed cron-scheduled pipelines for fault-tolerant, continuous updates and developed a command-line + dashboard interface, giving the engineering operations team real-time access to SOL-based queries, visualizations, and statistical summaries.

University of California - San Diego

Data Science Intern – Climate Systems

San Diego, CA, USA

October 2024 - May 2025

- Developed a forecasting algorithm for California coastal sea surface temperatures (SST) by applying statistical bias-correction techniques (diurnal adjustment matrices, regression-based scaling factors) to a century of observational data
- Developed downscaling models that applied a region specific sensitivity coefficient and global IPCC AR6 warming trajectories, producing station-level time series projections preserving seasonal periodicity while embedding long-term climate signals
- Built automated pipelines in Python and R for large-scale time-series analysis, anomaly detection, and visualization
- Validated outputs using residual analysis and uncertainty quantification, refining algorithms to minimize error while capturing nonlinear geographic gradients in SST warming
- My work increased my knowledge in transferable expertise within time-series modeling, statistical inference, and algorithm design.

National Oceanic and Atmospheric Administration (NOAA)

Data Engineering Intern – HPC & Cloud Systems

Remote

May 2024 - August 2024

- Contributed to advancing operational ocean forecasting by adapting the San Francisco Bay Operational Forecast System (SFBOFS) for NOAA's cloud sandbox, building transferable skills in cloud migration, distributed computing, and systems integration
- Partnered with applied mathematicians to design the first CDF-to-GRIB file format converter, improving the accessibility of high-volume model outputs
- Leveraged existing meshes and computational grids to reconfigure system code for cloud execution, gaining experience in translating HPC workflows to more flexible cloud architectures while maintaining forecast accuracy
- Conducted comparative analyses of global versus regional ocean model forcing, assessing impacts on water levels, currents, temperature, and salinity
- Produced performance benchmarks and technical insights that informed future strategies for NOAA's operational modeling efforts
- This role taught me to think like a systems engineer: validating model fidelity under new computational constraints, optimizing workflows for efficiency, and bridging scientific modeling with mission-critical operations.

University of California - San Diego

San Diego, CA, USA

Quantitative Analysis Intern – Climate Economics

May 2024 - August 2024

- Developed statistical models to quantify the social cost of carbon in marine ecosystems, writing Python/R scripts for large-scale data analysis and valuation of blue natural capital
- Designed methodologies for estimating economic benefits and biophysical damages of benthic ecosystems, incorporating regression analysis and sensitivity testing across species such as mangroves, corals, and deep-sea invertebrates
- Performed data centric assessments of climate change impacts on ocean-based markets including fisheries, aquaculture, and export goods, applying statistical inference and scenario-based forecasting to evaluate economic states
- Work bridged biophysical modeling with socio-economic valuation, strengthening my expertise in algorithm design, applied statistics, and quantitative analysis of complex systems.

University of California - San Diego

San Diego, CA, USA

Computational Modeling Intern – CCE-LTER

June 2023 - August 2023

- Developed algorithms and simulation models in MATLAB to analyze ecological responses of the California Current Ecosystem (CCE) to large-scale climate drivers, including El Niño and variable upwelling regimes
- Built 2D and 3D numerical models to evaluate environmental forcing on marine life dynamics, applying statistical methods and simulation frameworks to capture nonlinear ecosystem behavior
- Enhanced expertise in algorithm design, time-series modeling, and simulation-based forecasting, translating ecological questions into computational problems
- Work strengthened my skills in numerical analysis, scientific software development, and data interpretation.

The Molecular Sciences Software Institute

Los Angeles, CA, USA

Computational Modeling Intern

January 2023 - February 2023

- Designed computational molecular functions, with simulation software in Python and MATLAB with scientific libraries for large-scale data analysis and clustering
- Applied mathematical modeling techniques and algorithm design to optimize molecular simulations, integrating software with hardware components for improved performance and scalability
- Explored machine learning approaches to simulation fidelity, deployed workflows on NCSA's Polaris supercomputer
- Work emphasized translating mathematical models into production-ready code, strengthening my skills in numerical methods, parallel computation, and applied machine learning for scientific computing.

Skills

Skills: Python, NumPy, Pandas, Bash/Shell scripting, Statistical inference & regression analysis, Residual analysis, C/C++, Data pipeline design & real-time ingestion systems, Full-stack data systems, Git/GitHub version control, Sensitivity analysis & uncertainty quantification, MATLAB (signal processing, clustering, numerical methods), R (statistical modeling, data visualization), Machine Learning, Scikit-learn, Pytorch, Distributed computing & parallelization strategies

Languages: Spanish